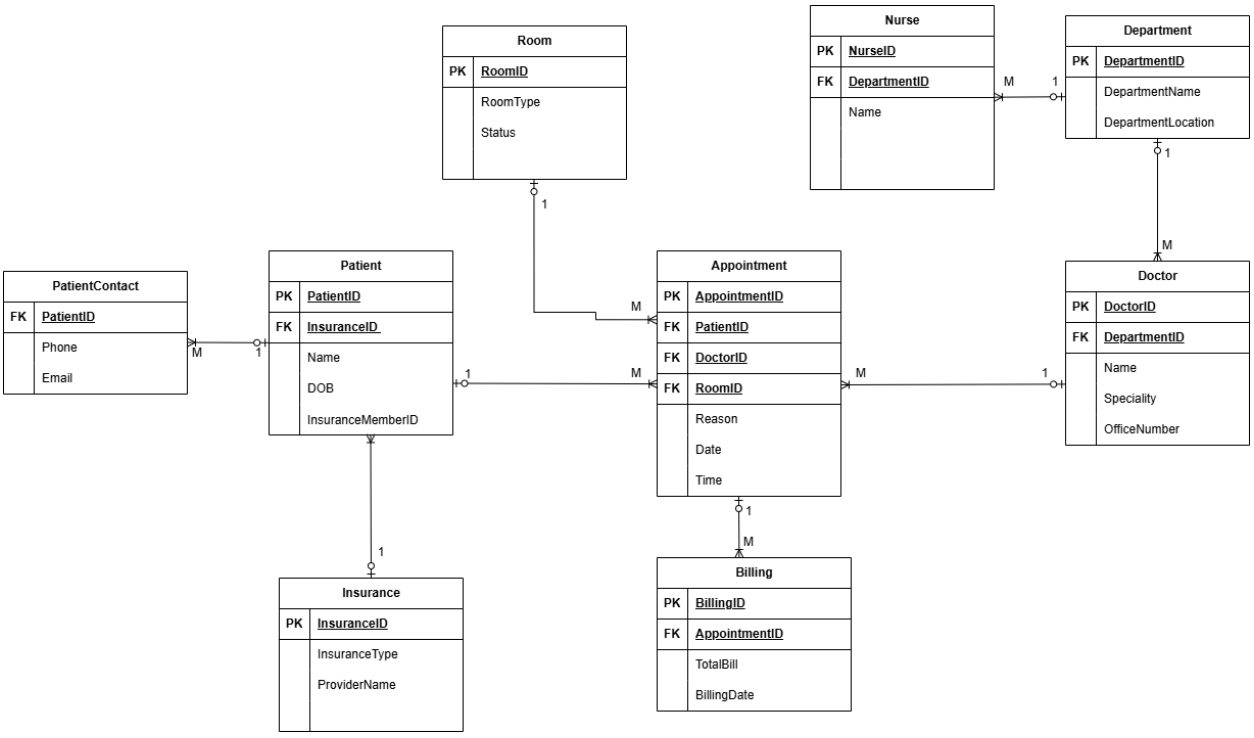
Hospital Database Management

The hospital currently manages patient data, doctor schedules, and department assignments through fragmented systems. This has led to redundant data entry and difficulty in tracking which doctors are available for specific appointments.

The goal of this database system is to centralize hospital operations into a structured relational database that supports efficient scheduling, patient management, department organization, billing, and insurance tracking.

The system ensures accurate data consistency, reduces duplication, and allows quick retrieval of medical and administrative information.

ERD



Business Rules

Based on the system requirements, here are the rules that will govern our database:

1. Each Doctor works in exactly one Department.
2. A Department can have many Doctors.
3. A Patient can have many Appointments.
4. Each Appointment is with exactly one Patient and exactly one Doctor.

5. A Doctor can have many Appointments.
6. A Patient may exist without having any appointments yet.
7. A Doctor may exist without having any appointments yet.
8. An Appointment cannot exist without both a Patient and a Doctor.
9. Each Appointment is assigned to exactly one Room.
10. A Room can host many appointments over time.
11. Each Nurse is assigned to exactly one Department.
12. A Department can have multiple Nurses.
13. A Patient may have zero or one Insurance provider.
14. One Insurance provider can cover many Patients.
15. Each Appointment can generate multiple Billing records.
16. Each Billing record must be linked to exactly one Appointment.

Normalization & Relational Schema

UNF

Management (PatientID, PatientName, Gender, DOB, Phone, Email, InsuranceType, ProviderName, InsuranceMemberID, DoctorID, DoctorName, Speciality, OfficeNumber, DepartmentName, DepartmentID, DepartmentLocation, AppointmentID, AppointmentReason, Date, Time, RoomID, RoomType, RoomStatus, BillingID, TotalBill, BillingDate, NurseID, NurseName)

1NF

Management(PatientID, PatientName, Gender, DOB, InsuranceType, ProviderName, InsuranceMemberID, DoctorID, DoctorName, Speciality, OfficeNumber, DepartmentID, DepartmentName, DepartmentLocation, AppointmentID, AppointmentReason, Date, Time, RoomID, RoomType, RoomStatus, BillingID, TotalBill, BillingDate, NurseID, NurseName)

PatientContact(PatientID, Phone, Email)

2NF

Patient(PatientID, PatientName, DOB, InsuranceType, ProviderName, Gender, InsuranceMemberID)

Doctor(DoctorID, DoctorName, Speciality, DepartmentID, OfficeNumber)

Department(DepartmentID, DepartmentName, DepartmentLocation)

Appointment(AppointmentID, AppointmentReason, Date, Time, PatientID, DoctorID, RoomID)

Room(RoomID, RoomType, RoomStatus)

Billing(BillingID, TotalBill, BillingDate, AppointmentID)

Nurse(NurseID, NurseName, DepartmentID)

PatientContact(PatientID, Phone, Email)

3NF

Patient(PatientID, PatientName, DOB, InsuranceID, Gender, InsuranceMemberID)

Insurance(InsuranceID, InsuranceType, ProviderName)

PatientContact(PatientID, Phone, Email)

Doctor(DoctorID, DoctorName, Speciality, DepartmentID, OfficeNumber)

Department(DepartmentID, DepartmentName, DepartmentLocation)

Nurse(NurseID, NurseName, DepartmentID)

Appointment(AppointmentID, AppointmentReason, Date, Time, PatientID, DoctorID, RoomID)

Room(RoomID, RoomType, RoomStatus)

Billing(BillingID, TotalBill, BillingDate, AppointmentID)

Key Designs

Room Allocation

Rooms are assigned based on medical need:

- ICU: critical/emergency cases
- Surgery: operations and procedures
- Radiology: diagnostic imaging
- Cardiac: heart-related cases
- Pediatric: child patients
- Emergency: urgent walk-ins
- General: routine consultations

Room status is managed dynamically during appointment scheduling to ensure no double-booking occurs.

Doctor Assignment

Doctors are assigned based on specialization and department:

- Cardiologists: Cardiac cases
- Surgeons: Surgical rooms
- Radiologists: Imaging diagnostics
- Emergency Specialists: Emergency cases
- Pediatricians: Pediatric cases

This ensures alignment between medical specialization and hospital resource usage.

Billing Design

Billing supports multiple charges per appointment through:

- Consultation fees
- Lab tests
- Surgery charges
- Medication
- Follow-ups

Each billing record is linked to exactly one appointment.

SQL Implementation

Database created and used

```
MariaDB [(none)]> CREATE DATABASE HospitalManagement;
Query OK, 1 row affected (0.002 sec)

MariaDB [(none)]> USE HospitalManagement;
Database changed
MariaDB [HospitalManagement]>
```

Insurance Table

```
MariaDB [(none)]> USE HospitalManagement;
Database changed
MariaDB [HospitalManagement]> CREATE TABLE Insurance (
    ->     InsuranceID VARCHAR(255) PRIMARY KEY NOT NULL,
    ->     ProviderName VARCHAR(255) NOT NULL,
    ->     InsuranceType VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.022 sec)
```

Department

```
MariaDB [HospitalManagement]> CREATE TABLE Department (
    ->     DepartmentID INT PRIMARY KEY AUTO_INCREMENT,
    ->     DepartmentName VARCHAR(255) NOT NULL,
    ->     DepartmentLocation VARCHAR(255) NOT NULL
    -> );
Query OK, 0 rows affected (0.022 sec)
```

Patient

```
MariaDB [HospitalManagement]> CREATE TABLE Patient (
    ->     PatientID INT PRIMARY KEY AUTO_INCREMENT,
    ->     PatientName VARCHAR(255) NOT NULL,
    ->     DOB DATE NOT NULL,
    ->     Gender VARCHAR(1),
    ->     InsuranceMemberID VARCHAR(255),
    ->     InsuranceID VARCHAR(255),
    ->     CONSTRAINT fk_patient_insurance
    ->     FOREIGN KEY (InsuranceID) REFERENCES Insurance(InsuranceID)
    -> );\
Query OK, 0 rows affected (0.022 sec)
```

Patient Contact

```
MariaDB [HospitalManagement]> CREATE TABLE PatientContact (
    ->     Phone VARCHAR(20) NOT NULL,
    ->     Email VARCHAR(50) NOT NULL,
    ->     PatientID INT NOT NULL,
    ->     CONSTRAINT fk_patient_phone
    ->     FOREIGN KEY (PatientID) REFERENCES Patient(PatientID)
    -> );
Query OK, 0 rows affected (0.025 sec)
```

Doctor

```
MariaDB [HospitalManagement]> CREATE TABLE Doctor (
    ->     DoctorID INT PRIMARY KEY AUTO_INCREMENT,
    ->     DoctorName VARCHAR(255) NOT NULL,
    ->     Speciality VARCHAR(255) NOT NULL,
    ->     OfficeNumber VARCHAR(255) NOT NULL,
    ->     DepartmentID INT,
    ->     CONSTRAINT fk_doctor_department
    ->     FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID)
    -> );
Query OK, 0 rows affected (0.008 sec)
```

Room

```
MariaDB [HospitalManagement]> CREATE TABLE Room (
    ->     RoomID INT PRIMARY KEY AUTO_INCREMENT,
    ->     RoomType VARCHAR(255) NOT NULL,
    ->     RoomStatus ENUM('Available', 'Occupied')
    -> );
Query OK, 0 rows affected (0.007 sec)
```

Appointment

```
MariaDB [HospitalManagement]> CREATE TABLE Appointment (
    ->     AppointmentID INT PRIMARY KEY AUTO_INCREMENT,
    ->     PatientID INT NOT NULL,
    ->     DoctorID INT NOT NULL,
    ->     AppointmentDate DATE NOT NULL,
    ->     AppointmentTime TIME NOT NULL,
    ->     Reason VARCHAR(255) NOT NULL,
    ->     RoomID INT,
    ->     CONSTRAINT fk_appointment_patient
    ->     FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),
    ->     CONSTRAINT fk_appointment_doctor
    ->     FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID),
    ->     CONSTRAINT fk_appointment_room
    ->     FOREIGN KEY (RoomID) REFERENCES Room(RoomID)
    -> );
Query OK, 0 rows affected (0.009 sec)
```

Nurse

```
MariaDB [HospitalManagement]> CREATE TABLE Nurse (
    ->     NurseID INT PRIMARY KEY AUTO_INCREMENT,
    ->     NurseName VARCHAR(255) NOT NULL,
    ->     DepartmentID INT,
    ->     CONSTRAINT fk_nurse_department
    ->     FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID)
    -> );
Query OK, 0 rows affected (0.008 sec)
```

Billing

```
MariaDB [HospitalManagement]> CREATE TABLE Billing (
    ->     BillingID INT PRIMARY KEY AUTO_INCREMENT,
    ->     TotalBill DECIMAL(10, 2) NOT NULL,
    ->     BillingDate DATE NOT NULL,
    ->     BillType VARCHAR(255) NOT NULL,
    ->     AppointmentID INT NOT NULL,
    ->     CONSTRAINT fk_billing_appointment
    ->     FOREIGN KEY (AppointmentID) REFERENCES Appointment(AppointmentID)
    -> );
Query OK, 0 rows affected (0.009 sec)
```

Insert Statements

Insurance

```
MariaDB [HospitalManagement]> INSERT INTO Insurance (InsuranceID, ProviderName, InsuranceType) VALUES
    -> ('BCBS-BASIC-01', 'Blue Cross Blue Shield', 'Basic'),
    -> ('BCBS-PREM-01', 'Blue Cross Blue Shield', 'Premium'),
    -> ('AETNA-BASIC-01', 'Aetna', 'Basic'),
    -> ('AETNA-PREM-01', 'Aetna', 'Premium'),
    -> ('CIGNA-BASIC-01', 'Cigna', 'Basic'),
    -> ('CIGNA-PREM-01', 'Cigna', 'Premium'),
    -> ('UHC-STD-01', 'UnitedHealthCare', 'Standard'),
    -> ('UHC-PREM-01', 'UnitedHealthCare', 'Premium'),
    -> ('HUMANA-STD-01', 'Humana', 'Standard'),
    -> ('HUMANA-PREM-01', 'Humana', 'Premium'),
    -> ('KAISER-STD-01', 'Kaiser', 'Standard'),
    -> ('KAISER-PREM-01', 'Kaiser', 'Premium');
Query OK, 12 rows affected (0.008 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

Department

```
MariaDB [HospitalManagement]> INSERT INTO Department (DepartmentName, DepartmentLocation) VALUES
    -> ('Cardiology', 'Building A'),
    -> ('Anesthesiology', 'Building B'),
    -> ('ENT', 'Building C'),
    -> ('Geriatrics', 'Building D'),
    -> ('Pediatrics', 'Building E'),
    -> ('Neurology', 'Building F'),
    -> ('Oncology', 'Building G'),
    -> ('Orthopedics', 'Building H'),
    -> ('Radiology', 'Building I'),
    -> ('Surgery', 'Building J'),
    -> ('Emergency', 'Building K');
Query OK, 11 rows affected (0.003 sec)
Records: 11  Duplicates: 0  Warnings: 0
```

Patient

```
MariaDB [HospitalManagement]> INSERT INTO Patient (PatientName, DOB, Gender, InsuranceID) VALUES
    -> ('John Webb', '1980-12-01', 'M', 'BCBS-PREM-01'),
    -> ('Sarah Jones', '1995-09-23', 'F', 'BCBS-BASIC-01'),
    -> ('Ron Brown', '2005-01-15', 'M', 'AETNA-BASIC-01'),
    -> ('Emily Wilson', '2004-07-30', 'F', 'AETNA-PREM-01'),
    -> ('William Johnson', '1988-11-10', 'M', 'CIGNA-PREM-01'),
    -> ('Lily Martin', '1990-03-05', 'F', 'CIGNA-BASIC-01'),
    -> ('James Miller', '1985-02-28', 'M', 'UHC-PREM-01'),
    -> ('Sophia Anderson', '1992-06-18', 'F', 'KAISER-PREM-01');
Query OK, 8 rows affected (0.002 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

PatientContact

```
MariaDB [HospitalManagement]> INSERT INTO PatientContact (Phone, Email, PatientID) VALUES
    -> ('123-456-7890', 'john.webb@gmail.com', 9),
    -> ('234-567-8901', 'sarah.jones@gmail.com', 10),
    -> ('345-678-9012', 'ron.brown@gmail.com', 11),
    -> ('456-789-0123', 'emily.wilson@gmail.com', 12),
    -> ('567-890-1234', 'william.johnson@gmail.com', 13),
    -> ('678-901-2345', 'lily.martin@gmail.com', 14),
    -> ('789-012-3456', 'james.miller@gmail.com', 15),
    -> ('890-123-4567', 'sophia.anderson@gmail.com', 16);
Query OK, 8 rows affected (0.013 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Doctor

```
MariaDB [HospitalManagement]> INSERT INTO Doctor (DoctorName, Speciality, OfficeNumber, DepartmentID) VALUES
    -> ('Dr. Smith', 'Cardiologist', '101', 1),
    -> ('Dr. Harris', 'Cardiologist', '112', 1),
    -> ('Dr. Johnson', 'Anesthesiologist', '102', 2),
    -> ('Dr. Lee', 'ENT Specialist', '103', 3),
    -> ('Dr. Brown', 'Geriatrician', '104', 4),
    -> ('Dr. Davis', 'Pediatrician', '105', 5),
    -> ('Dr. Wilson', 'Neurologist', '106', 6),
    -> ('Dr. Taylor', 'Oncologist', '107', 7),
    -> ('Dr. Anderson', 'Orthopedic Surgeon', '108', 8),
    -> ('Dr. Thomas', 'Radiologist', '109', 9),
    -> ('Dr. Jackson', 'Surgeon', '110', 10),
    -> ('Dr. White', 'Emergency Specialist', '111', 11);
Query OK, 12 rows affected (0.015 sec)
Records: 12  Duplicates: 0  Warnings: 0
```

Room

```
MariaDB [HospitalManagement]> INSERT INTO Room (RoomType, RoomStatus) VALUES
    -> ('General', 'Available'),
    -> ('ICU', 'Occupied'),
    -> ('Emergency', 'Occupied'),
    -> ('Recovery', 'Available'),
    -> ('Surgery', 'Available'),
    -> ('Radiology', 'Available'),
    -> ('Pediatric', 'Available'),
    -> ('Cardiac', 'Occupied');
Query OK, 8 rows affected (0.015 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Appointment

```
MariaDB [HospitalManagement]> INSERT INTO Appointment (PatientID, DoctorID, AppointmentDate, AppointmentTime, Reason, RoomID) VALUES
    -> (9, 1, '2026-04-10', '09:00:00', 'Chest pain', 8),
    -> (10, 2, '2026-04-11', '10:00:00', 'Surgery consultation', 5),
    -> (11, 3, '2026-04-12', '11:00:00', 'Stomach pain', 3),
    -> (12, 4, '2026-04-13', '12:00:00', 'Regular checkup', 1),
    -> (13, 5, '2026-04-14', '13:00:00', 'Child fever', 7),
    -> (14, 6, '2026-04-15', '14:00:00', 'Headache scan', 6),
    -> (15, 7, '2026-04-16', '15:00:00', 'Cancer diagnosis', 2),
    -> (16, 8, '2026-04-17', '16:00:00', 'Bone fracture treatment', 5);
Query OK, 8 rows affected (0.005 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Nurse

```
MariaDB [HospitalManagement]> INSERT INTO Nurse (NurseName, DepartmentID) VALUES
    -> ('Nurse Anna', 1),
    -> ('Nurse Betty', 2),
    -> ('Nurse Carol', 3),
    -> ('Nurse Diana', 4),
    -> ('Nurse Emily', 5),
    -> ('Nurse Frank', 6),
    -> ('Nurse Gianna', 7),
    -> ('Nurse Hailey', 8),
    -> ('Nurse Isabella', 9),
    -> ('Nurse Jane', 10),
    -> ('Nurse Karen', 11);
Query OK, 11 rows affected (0.013 sec)
Records: 11  Duplicates: 0  Warnings: 0
```

Billing

```
MariaDB [HospitalManagement]> INSERT INTO Billing (TotalBill, BillingDate, BillType, AppointmentID) VALUES
    -> (500.00, '2026-04-10', 'Consultation', 1),
    -> (100.00, '2026-04-10', 'Lab Test', 1),
    -> (1500.00, '2026-04-11', 'Surgery Fee', 2),
    -> (200.00, '2026-04-12', 'Medication', 3),
    -> (300.00, '2026-04-14', 'Follow-up', 5);
Query OK, 5 rows affected (0.015 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

JOIN- shows which insurance each patient has

```
MariaDB [HospitalManagement]> SELECT p.PatientName,p.Gender,i.ProviderName,i.InsuranceType FROM Patient p
    -> JOIN Insurance i ON p.InsuranceID = i.InsuranceID;
+-----------------+--------+-----------------------+---------------+
| PatientName     | Gender | ProviderName          | InsuranceType |
+-----------------+--------+-----------------------+---------------+
| John Webb       | M      | Blue Cross Blue Shield | Premium       |
| Sarah Jones     | F      | Blue Cross Blue Shield | Basic         |
| Ron Brown       | M      | Aetna                 | Basic         |
| Emily Wilson    | F      | Aetna                 | Premium       |
| William Johnson | M      | Cigna                 | Premium       |
| Lily Martin     | F      | Cigna                 | Basic         |
| James Miller    | M      | UnitedHealthCare      | Premium       |
| Sophia Anderson | F      | Kaiser                | Premium       |
+-----------------+--------+-----------------------+---------------+
8 rows in set (0.001 sec)
```

LEFT JOIN- Patient without insurance (empty set in this case)

```
MariaDB [HospitalManagement]> SELECT p.PatientName,p.Gender,i.ProviderName,i.InsuranceType FROM Patient p
    -> JOIN Insurance i ON p.InsuranceID = i.InsuranceID WHERE i.InsuranceID IS NULL;
Empty set (0.013 sec)
```

Doctors with their department

```
MariaDB [HospitalManagement]> SELECT d.DoctorName, d.Speciality, dept.DepartmentName FROM Doctor d
    -> JOIN Department dept ON d.DepartmentID = dept.DepartmentID;
+--------------+----------------------+----------------+
| DoctorName   | Speciality           | DepartmentName |
+--------------+----------------------+----------------+
| Dr. Smith    | Cardiologist         | Cardiology     |
| Dr. Harris   | Cardiologist         | Cardiology     |
| Dr. Johnson  | Anesthesiologist     | Anesthesiology |
| Dr. Lee      | ENT Specialist       | ENT            |
| Dr. Brown    | Geriatrician         | Geriatrics     |
| Dr. Davis    | Pediatrician         | Pediatrics     |
| Dr. Wilson   | Neurologist          | Neurology      |
| Dr. Taylor   | Oncologist           | Oncology       |
| Dr. Anderson | Orthopedic Surgeon   | Orthopedics    |
| Dr. Thomas   | Radiologist          | Radiology      |
| Dr. Jackson  | Surgeon              | Surgery        |
| Dr. White    | Emergency Specialist | Emergency      |
+--------------+----------------------+----------------+
12 rows in set (0.006 sec)
```

Appointment with Patient and Doctor

```
MariaDB [HospitalManagement]> SELECT app.AppointmentDate, app.AppointmentTime, p.PatientName, d.DoctorName, app.Reason FROM Appointment app
    -> JOIN Patient p ON app.PatientID = p.PatientID
    -> JOIN Doctor d ON app.DoctorID = d.DoctorID;
+-----------------+-----------------+---------------+------------+------------------------+
| AppointmentDate | AppointmentTime | PatientName   | DoctorName | Reason                 |
+-----------------+-----------------+---------------+------------+------------------------+
| 2026-04-10      | 09:00:00        | John Webb     | Dr. Smith  | Chest pain             |
| 2026-04-11      | 10:00:00        | Sarah Jones   | Dr. Harris | Surgery consultation   |
| 2026-04-12      | 11:00:00        | Ron Brown     | Dr. Johnson| Stomach pain           |
| 2026-04-13      | 12:00:00        | Emily Wilson  | Dr. Lee    | Regular checkup        |
| 2026-04-14      | 13:00:00        | William Johnson| Dr. Brown | Child fever            |
| 2026-04-15      | 14:00:00        | Lily Martin   | Dr. Davis  | Headache scan          |
| 2026-04-16      | 15:00:00        | James Miller  | Dr. Wilson | Cancer diagnosis       |
| 2026-04-17      | 16:00:00        | Sophia Anderson| Dr. Taylor| Bone fracture treatment|
+-----------------+-----------------+---------------+------------+------------------------+
8 rows in set (0.013 sec)
```

Appointments per Doctor

```
MariaDB [HospitalManagement]> SELECT d.DoctorName, COUNT(app.AppointmentID) AS TotalAppointments
    -> FROM Doctor d
    -> LEFT JOIN Appointment app
    -> ON d.DoctorID = app.DoctorID
    -> GROUP BY d.DoctorName;
+--------------+--------------------+
| DoctorName   | TotalAppointments  |
+--------------+--------------------+
| Dr. Anderson |                  0 |
| Dr. Brown    |                  1 |
| Dr. Davis    |                  1 |
| Dr. Harris   |                  1 |
| Dr. Jackson  |                  0 |
| Dr. Johnson  |                  1 |
| Dr. Lee      |                  1 |
| Dr. Smith    |                  1 |
| Dr. Taylor   |                  1 |
| Dr. Thomas   |                  0 |
| Dr. White    |                  0 |
| Dr. Wilson   |                  1 |
+--------------+--------------------+
12 rows in set (0.002 sec)
```

Total Bill per appointment

```
MariaDB [HospitalManagement]> SELECT AppointmentID, SUM(TotalBill) AS TotalBill FROM Billing
    -> GROUP BY AppointmentID;
+---------------+-----------+
| AppointmentID | TotalBill |
+---------------+-----------+
|             1 |    600.00 |
|             2 |   1500.00 |
|             3 |    200.00 |
|             5 |    300.00 |
+---------------+-----------+
4 rows in set (0.001 sec)
```